

Name: _____

Class: _____

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12

HSC ASSESSMENT TASK 3

JUNE 2010

MATHEMATICS Extension 1

Time Allowed: 70 minutes

Instructions:

- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start **each** question on a **new page**.
- Standard integrals can be found on the last page.

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total
/9	/8	/9	/8	/8	/8	/50

Question 1

- a) Write the exact value of :
- i) $\cos^{-1}(\cos \frac{3\pi}{2})$ 1
 - ii) $\sin^{-1}0.4 + \cos^{-1}0.4$ 1
- b) Simplify $\frac{\sin(\pi-x)}{\sin(\frac{\pi}{2}-x)}$ 2
- c) Evaluate $\log_9 30$ correct to 2 decimal places. 1
- d) Solve, leaving answers in exact form:
- i) $3^{x-1} = 7$ 2
 - ii) $\ln(x^2) + \ln x = 1$ 2

Question 2

- a) Evaluate $\lim_{x \rightarrow 0} \frac{\frac{x}{3}}{\sin 2x}$ 1
- b) Differentiate:
- i) $\tan 2x$ 1
 - ii) $\ln(\frac{x}{x^2+3})$ 2
 - iii) $\operatorname{cosec}^2 x$ 2
 - iv) $\sin^{-1} 3x$ 2

Question 3

- a) Find:
- i) $\int e^{4x} dx$ 1
 - ii) $\int \frac{1+2x}{x^2} dx$ 2
 - iii) $\int \frac{x}{1+2x^2} dx$ 1

- b) Using the substitution $u = \tan x$, or otherwise, find $\int \sec^2 x \tan^2 x \, dx$. 2
- c) i) Find $\frac{d}{dx}(xe^{2x})$ 1
- ii) Hence or otherwise, find $\int xe^{2x} \, dx$ 2

Question 4

- a) On the same axes, sketch $y = \sin x$ and $y = \ln(\sin x)$ for $0 \leq x \leq \pi$. 2
Clearly label key features.
- b) i) Express $\sin^2 x$ in terms of $\cos 2x$ 1
- ii) The curve $y = \sin 2x$, for $0 \leq x \leq \pi$, is rotated about the x axis. 3
Find the total volume generated.
- c) Evaluate $\sin\left[\tan^{-1}\left(\frac{5}{4}\right)\right]$ in exact form. 2

Question 5

A function f is defined $f(x) = x^2 - 2x$.

- a) State the largest positive domain for f to have an inverse function f^{-1} . 1
- b) State the domain and range of f^{-1} . 2
- c) Sketch f and f^{-1} on the same axes for the domains and ranges above. 2
Clearly show key points.
- d) Find the inverse function $f^{-1}(x)$. 2
- e) Find the value of x for which $f(x) = f^{-1}(x)$. 1

Question 6

- a) Express $\cos^{-1}\left(\frac{1}{3}\right) + \cos^{-1}\left(\frac{1}{4}\right)$ in the form $\cos^{-1}M$. 2
- b) i) Write the domain and range for $y = 3\sin^{-1}\left(\frac{x}{2}\right)$ 2
- ii) Sketch the curve in i) 1
- iii) Evaluate $\int_0^1 3\sin^{-1}\left(\frac{x}{2}\right) \, dx$. 3
Leave your answer in exact form.

SOLUTIONS

① a) i) $\cos^{-1} 0 = \frac{\pi}{2}$

ii) $\frac{\pi}{2}$

b) $\frac{\sin x}{\cos x} = \tan x$

c) $\frac{\log 30}{\log 9} \doteq 1.55$

d) $\log(3^{x-1}) = \log 7$
 $(x-1) \log 3 = \log 7$
 $x \log 3 - \log 3 = \log 7$
 $x = \frac{\log 7 + \log 3}{\log 3}$

e) $\log_e(x^3) = 1$

$\therefore x^3 = e^1$

$\therefore x = \sqrt[3]{e}$

② a) $\lim_{x \rightarrow 0} \frac{2x}{\sin 2x} \times \frac{1}{6} = 1 \times \frac{1}{6}$
 $= \frac{1}{6}$ ①

b) i) $2 \sec^2 2x$ ①

ii) $\log x - \log(x^2 + 3)$
 $\frac{dy}{dx} = \frac{1}{x} - \frac{2x}{x^2 + 3}$ ②

iii) $y = \frac{1}{\sin^2 x}$
 $\frac{dy}{dx} = \frac{0 - 2 \sin x \cos x}{\sin^4 x}$

②
 $= \frac{-2 \cos x}{\sin^3 x} \text{ or } -2 \cot x \operatorname{cosec}^2 x$

iv) $\frac{dy}{dx} = \frac{1}{\sqrt{1-(3x)^2}} \times 3$
 $= \frac{3}{\sqrt{1-9x^2}}$ ②

③ a) i) $\frac{e^{4x}}{4} + c$

ii) $\int \left(\frac{1}{x^2} + \frac{2x}{x^2} \right) dx$
 $= \int (x^{-2} + \frac{2}{x}) dx$
 $= -x^{-1} + 2 \log x + c$
 $= -\frac{1}{x} + 2 \log x + c$

iii) $\frac{1}{4} \int \frac{4x}{1+2x^2} dx$
 $= \frac{1}{4} \log(1+2x^2) + c$

b) $\int \sec^2 x \tan^2 x dx = \int \sec^2 x u^2 \frac{du}{\sec^2 x}$

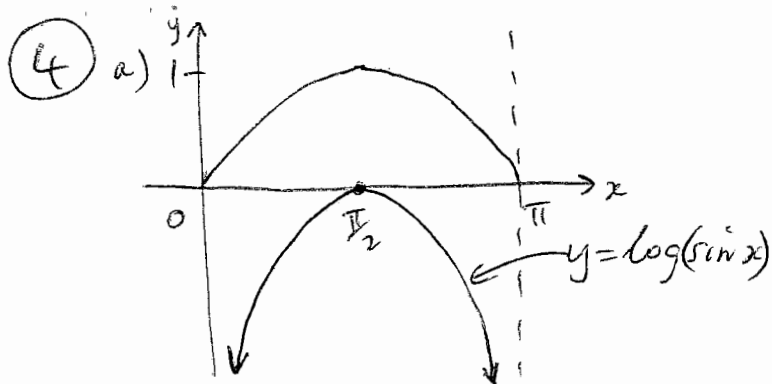
$u = \tan x$
 $\frac{du}{dx} = \sec^2 x$
 $dx = \frac{du}{\sec^2 x}$

$= \frac{u^3}{3} + c$
 $= \frac{\tan^3 x}{3} + c$

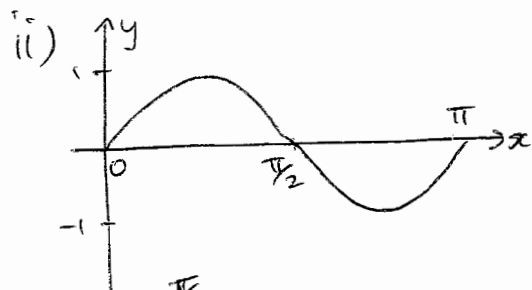
c) i) $\frac{d}{dx}(xe^{2x}) = e^{2x} + 2xe^{2x}$

ii) $2xe^{2x} = \frac{d}{dx}(xe^{2x}) - e^{2x}$

$\therefore \int xe^{2x} = \frac{1}{2} \int \frac{d}{dx}(xe^{2x}) - \frac{1}{2} \int e^{2x} dx$
 $= \frac{1}{2} xe^{2x} - \frac{1}{4} e^{2x} + c$



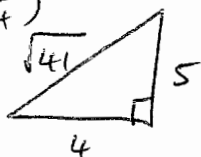
b) i) $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$



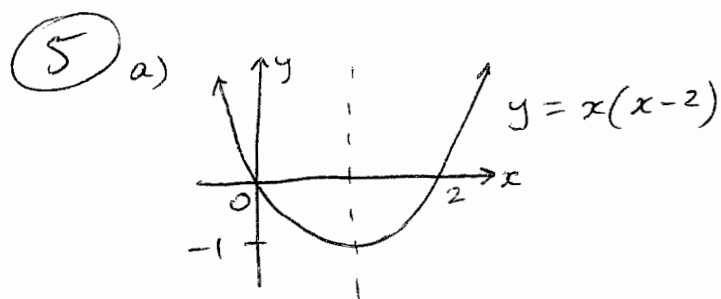
$$\begin{aligned}
 \text{Vol} &= 2\pi \int_0^{\pi/2} \sin^2 2x \, dx \\
 &= 2\pi \int_0^{\pi/2} \frac{1}{2}(1 - \cos 4x) \, dx \\
 &= \pi \left[x - \frac{\sin 4x}{4} \right]_0^{\pi/2} \\
 &= \pi \left[\left(\frac{\pi}{2} - 0 \right) - (0 - 0) \right] \\
 &= \frac{\pi^2}{2} \text{ units}^3.
 \end{aligned}$$

c) Let $\alpha = \tan^{-1}\left(\frac{5}{4}\right)$

$\therefore \tan \alpha = \frac{5}{4}$

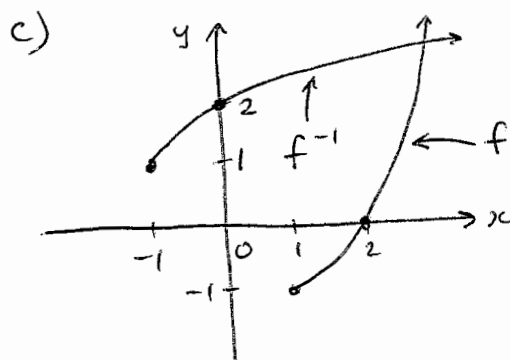


$$\begin{aligned}
 \therefore \sin\left(\tan^{-1}\frac{5}{4}\right) &= \sin \alpha \\
 &= \frac{5}{\sqrt{41}}
 \end{aligned}$$



$\therefore$ domain: $x \geq 1$

b) For f^{-1} , $D: x \geq -1$
 $R: y \geq 1$



d) $f^{-1}(x) \Rightarrow x = y^2 - 2y$

$$y^2 - 2y + 1 = x + 1$$

$$(y-1)^2 = x+1$$

$$y-1 = \sqrt{x+1} \text{ only}$$

$$y = \sqrt{x+1} + 1$$

e) graphs intersect on $y=x$

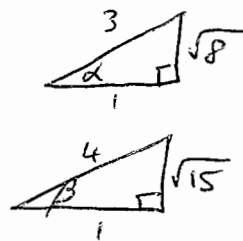
$\therefore$ solve $x^2 - 2x = x$

$$x^2 - 3x = 0$$

$$x(x-3) = 0$$

$\therefore \underline{x=3 \text{ only}}$ ($x=0$ not applicable).

⑥ a) Let $\alpha = \cos^{-1}(\frac{1}{3}) \Rightarrow \cos \alpha = \frac{1}{3}$
 $\beta = \cos^{-1}(\frac{1}{4}) \Rightarrow \cos \beta = \frac{1}{4}$



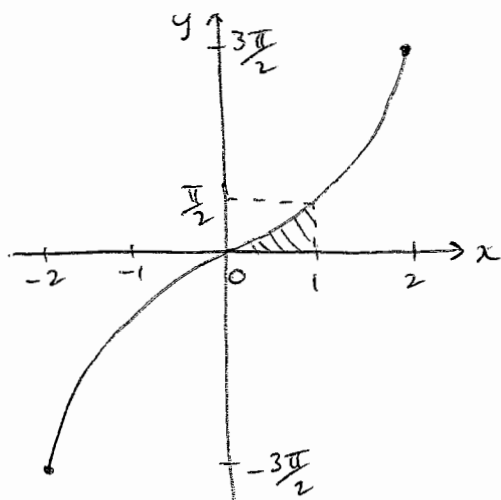
$$\begin{aligned}\cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= \frac{1}{3} \times \frac{1}{4} - \frac{\sqrt{8}}{3} \times \frac{\sqrt{15}}{4} \\ &= \frac{1}{12} - \frac{\sqrt{120}}{12} \\ &= \frac{1 - \sqrt{120}}{12}\end{aligned}$$

$\therefore \alpha + \beta = \cos^{-1}\left(\frac{1 - \sqrt{120}}{12}\right)$ as reqd.

b) i) D : $-2 \leq x \leq 2$

R : $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

ii)



iii) Shaded area = rectangle - $\int_0^{\pi/2} 2 \sin(\frac{y}{3}) dy$

$$\begin{aligned}\therefore \int_0^1 3 \sin^{-1}\left(\frac{x}{2}\right) dx &= \frac{\pi}{2} - \left[-2 \cos\left(\frac{y}{3}\right) \times 3 \right]_0^{\pi/2} \\ &= \frac{\pi}{2} + 6 \left[\cos \frac{y}{3} \right]_0^{\pi/2} \\ &= \frac{\pi}{2} + 6 \left(\frac{\sqrt{3}}{2} - 1 \right) \\ &= \frac{\pi}{2} + 3\sqrt{3} - 6\end{aligned}$$